

PAH-OMC-029: Uniqueness of the fixed Markov extension

TECT verification-first repository
claim C6-SPACETIME-SIGNATURE

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Purpose and scope

This is the single synthesis checkpoint for T-096 / R-568. The exact preregistration, full proof and independent commutator audit are PAH-OMC-029-prereg-v1.json, PAH-OMC-029-certificate.md and PAH-OMC-029-independent-audit.md, under strategy/pa-hyp/. No PAH source is modified.

Use the R-510 labelled all- Q probability law μ and its real gauge-invariant space $H = L^2(\mu)_{\text{inv}}$. Let D be exactly R-511's bounded, globally amplitude- ℓ^1 -Lipschitz, invariant finite-prefix cylinders and A its PH/LK/AP generator. Retain the half-energy exponent, coincident directed channels and partial AP domains. The amplitudes do not move under A . The scope is the anchored two-row strip with $K = 2$, $M_s = 1$, $\epsilon = 1/2$, $\beta = \nu = 1$, $m^2 = \theta = 0$, other OMC-016 couplings one. The previously specified j -before-anchored- n state construction is inherited, not repeated or exchanged.

Precise conditional theorem

Let $S(t)$ be any symmetric, strongly continuous contraction semigroup on this same H , positive and L^∞ -contractive, with generator G satisfying $D \subset \text{Dom}(G)$ and $G|_D = A$. Then

$$S(t) = T_{\min}(t) \quad (t \geq 0),$$

where $T_{\min}$ is the R-512 extension with its Markov proof repaired in R-567. No minimal form domain, graph core, path realization or boundary condition is imposed on the competitors. This is Markov-extension uniqueness, not essential self-adjointness and not finite PAH convergence.

Every extension respects radial localization

For radial $a \in D$, $Ga = Aa = 0$. Generator closedness and density of radial cylinders show that the full radial subspace H_{rad} lies in $\ker G$. Thus $S(t)a = a$ there, including $a = 1$. For any radial event E and $0 \leq f \leq M$, positivity gives

$$0 \leq S(t)(1_E f) \leq MS(t)1_E = M1_E.$$

The complementary event gives a disjoint image. By additivity and bounded approximation, $S(t)1_E f = 1_E S(t)f$ for all $f \in H$. Differentiation at zero therefore gives $G(1_E f) = 1_E Gf$ for $f \in \text{Dom}(G)$. This is an all-extension statement; it uses no infinite process.

Uniform state tail and the exact rate envelope

Write K for the nonsymmetric SPATIAL transfer of R-510, not the time generator. Its source bounds are $K(x, y) \leq 8u(x)u(y)$ and $u^2 \leq \exp[-(r_0^6 + r_1^6)/6]$. With its normalized positive right eigenfunction ϕ , set

$U = \|u\|_2$, $c = \langle u, \phi \rangle > 0$ and $B = \|P\| + C$, where $\|\lambda^{-i}K^i - P\| \leq Cq^i$. The anchored column density, not a translation-invariant surrogate, is

$$\rho_i = \frac{(K^*)^i u \phi}{\lambda^i c}.$$

The product envelope gives $\phi \leq 8Uu/\lambda$ and, for $i \geq 1$, $(K^*)^i u \leq 8B\lambda^{i-1}U^2u$. Treat $i = 0$ separately. Hence

$$\rho_i \leq D_0 u^2, \quad D_0 = \max\left(\frac{8U}{\lambda c}, \frac{64BU^3}{\lambda^2 c}\right).$$

It follows that $\sup_{i,v} \mu(e^{r_{i,v}^6/12}) \leq C_{\text{tail}} < \infty$. These are finite operator-defined constants, not numerical spectral-gap certificates or estimates uniform in changed source parameters.

Set $\gamma = 1/12$, $a = 3/\gamma$ and

$$E_M = \{r_{i,v}^6 \leq M + a \log(i+2) \text{ for every } i, v\}.$$

Then $\mu(E_M^c) \leq 2C_{\text{tail}}e^{-\gamma M} \sum_{i \geq 0} (i+2)^{-3}$, which tends to zero. The original measure is not conditioned or renormalized.

For a root whose neighboring amplitudes are at most R , the displayed functional gives

$$|\Delta_r F| \leq a_0 + a_2 R^2, \quad a_0 = \frac{163}{4}, \quad a_2 = \frac{163}{8}.$$

The values are independently derived from

$$a_0 = (1+d)(1-\epsilon)^2/2 + 2d(2/\epsilon), \quad a_2 = (1-\epsilon^2)/2 + d(2/\epsilon), \quad d = 5.$$

Quartic and sextic onsite terms cancel in every label-only increment. At most d edges and $2d$ faces are affected. Each face lies in $[0, 2/\epsilon]$, each matter-edge term in $[0, 2R^2/\epsilon]$. The original mobility is at most one, so $c_r \leq e^{(a_0+a_2R^2)/2}$.

Assign forward links to their left column. There are $b = 8$ bits and $p = 16$ potential directed roots per column. Use the conservative dependence radius $R_c = 2$; the incidence proof gives radius at most one. A backward step has at most $B_c = 2b(2R_c + 1) = 80$ choices. On E_M the roots needed through column $N + R_c$ have rate bounded by

$$q_N = \exp\left\{\frac{a_0 + a_2[M + a \log(N + 2R_c + 2)]^{1/3}}{2}\right\}.$$

For each fixed M and every $\delta > 0$, $q_N \leq C_{M,\delta}(N + 2R_c + 2)^\delta$. This follows by maximizing $by - \delta y^3$; we use $\delta = 1/2$. No global cutoff or time rescaling is inserted.

The all-extension domain bridge

Let $A^{[N]}$ sum the exact roots centered in columns $0, \dots, N$, retaining their full exterior-dependent rates. Its finite label matrix defines $P_N(s)$ at each fixed amplitude vector and exterior labels. This is a backward test, not a substituted finite-volume Hamiltonian.

For $f \in D$ supported through $m \leq N$, $P_N(s)f$ reads only columns through $N + R_c$. It is gauge invariant and Lipschitz on every compact amplitude box by finite matrix-exponential regularity. A radial cutoff equal to one on the E_M prefix box makes it an element of the ORIGINAL D . The cutoff has zero A increment. Radial multiplier commutation therefore yields, for every candidate generator,

$$v_s = 1_{E_M} P_N(s)f \in \text{Dom}(G), \quad Gv_s = 1_{E_M} A P_N(s)f, \quad v'_s = 1_{E_M} A^{[N]} P_N(s)f.$$

These functions and derivatives are graph-norm continuous in s on compact intervals, using the finite number of bounded rates on that box. Neither $A(D) \subset D$ nor a new common graph core is assumed.

Boundary influence and uniqueness

For fixed T , put $d_N = \lceil (N + 1 - m)/R_c \rceil$, $x_N = B_c q_N T$ and $k_f = b(m + 1)$. A finite rate- q_N proposal construction, preserving every channel and its acceptance probability c_r/q_N , bounds exterior influence:

$$|\Delta_r P_N(s)f| \leq 2\|f\|_\infty k_f \sum_{k \geq d_N} \frac{x_N^k}{k!} \leq 2\|f\|_\infty k_f e^{x_N} \frac{x_N^{d_N}}{d_N!}.$$

Ordered proposals contribute $s^k/k!$ even when paths share channels; the proof uses a union bound, not independence between paths. At most pR_c exterior roots can change the test, so

$$\sup_{s \leq T} \|1_{E_M}(A - A^{[N]})P_N(s)f\|_H \leq 2\|f\|_\infty k_f pR_c q_N e^{x_N} \frac{x_N^{d_N}}{d_N!} =: \varepsilon_N.$$

Using $q_N = O(\sqrt{N})$ and $d! \geq (d/e)^d$ gives

$$\log \varepsilon_N \leq -\frac{N}{2R_c} \log N + O(N) + O(\sqrt{N}) \longrightarrow -\infty.$$

The constants depend on M, T, f , not N or the extension. The zero-vector and zero-time cases are immediate. Differentiating $S(t - s)v_s$ now gives

$$\sup_{t \leq T} \|S(t)1_{E_M}f - 1_{E_M}P_N(t)f\|_H \leq T\varepsilon_N.$$

Two arbitrary extensions share the same comparison. First take $N \rightarrow \infty$ at fixed M , then remove E_M using the remaining norm error $2\|f\|_\infty \sqrt{\mu(E_M^c)}$. Density of D proves equality on H . R-567 supplies one member, the minimal extension.

Independent formulation and verification coverage

The separate audit replaces proposals by a commutator argument. Write $L_N = \sum_j b_j(\tau_j - I)$, summing rather than deleting coincident channels. Since the binary flips commute, $w_i = (\tau_i - I)P_N(t)f$ satisfies

$$w'_i = L_N w_i + \sum_j (\tau_i b_j - b_j) \tau_i (\tau_j - I) P_N(t) f.$$

Finite variation of constants bounds oscillations by e^{tC} , with $C_{ij} = \|\tau_i b_j - b_j\|_\infty \leq 2q_N$ and row sum at most $B_c q_N$. Matrix powers below distance d_N vanish, giving the same factorial tail. This independently verifies the boundary estimate without an infinite graphical realization.

Primary and non-importing independent scripts recompute the source coefficients and root counts; hostile controls test multiplicity, half exponents, summability and the strict sublinear threshold. Six Lean theorems compile: closed-graph radial kernel, common comparison, scalar error removal, exact arithmetic, negative log coefficient, and cubic optimization. The measure/spectral inputs, complete domain argument, commutator comparison and full asymptotic theorem remain analytic, not fully formalized. Internal alternative derivations are not an external referee report. External review of the domain bridge is invited.

Devil's-advocate

1. Conservation or a minimal form alone proves uniqueness: UPHELD against the shortcut; the localization and boundary estimates are essential.

2. $P_N(s)f$ automatically belongs to D : UPHELD without the compact radial cutoff and all-extension multiplier proof.
3. Exterior bits can be dropped from rates: UPHELD against that change; they remain parameters and produce the controlled residual.
4. A globally bounded rate or fixed $R_{\max}$ is needed: DISMISSED here by the state-derived tempered events and subpolynomial rate envelope.
5. Linear rate growth is covered by the same decay argument: UPHELD against that inference; the logarithmic coefficient requires $\delta < 1$.
6. Uniqueness identifies original finite PAH dynamics: UPHELD against promotion. No finite-to-common-space realization or convergence is supplied.

Result footer

Result ID: R-568; contract PAH-OMC-029; task T-096
 Claim ID: C6-SPACETIME-SIGNATURE
 Precise statement: Every symmetric strongly continuous Markov semigroup on the same H whose generator extends A on D equals the R-512/R-567 minimal extension, under the named source inputs.
 Scope: Fixed R-510 state, R-511 domain and unchanged PH/LK/AP rates.
 Dependencies: R-510, R-511, R-512 and R-567; exact source hashes.
 Evidence grade: ANALYTIC, CONDITIONAL, EXACT, EXECUTED; partial Lean.
 Reproduction command: `python -X utf8 verification/scripts/pah_omc029_verify.py --check`
 Expected output: PAH-OMC-029 CHECKS PASS; six Lean theorems.
 Falsification gate: Invalid radial/domain or boundary estimate; two distinct admissible extensions under the same hypotheses.
 Tier before / after: T1 / T1; auxiliary_support; T-054 unchanged.
 No-overclaim statement: No original finite-semigroup convergence, Pre-A, spacetime, QFT, gravity, continuum, mass gap or TOE claim.
 Next required action: Source-authorized finite-to-common-space realization with limit generator extending this same A on D .